Amplifier Configuration

circuit -1

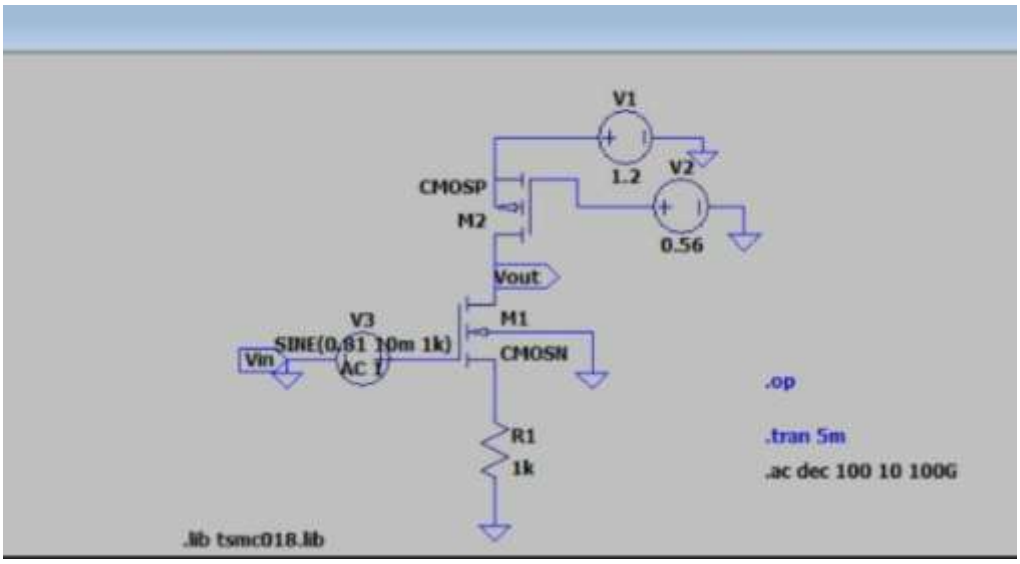
Aim:

To perform the DC, Transient, AC Analysis of a Common Source amplifier circuit using LTSpice and extract the associated parameters.

Components Required:

TSMC 180nm NMOS transistor (CMOSN), PMOS transistor (CMOSP),source resistor and voltage sources(3)

Circuit Diagram:



given parameters:

Vov=0.25v,Vth=0.36(NMOS),Vth (PMOS)=-0.39V,VDD=1.2,P<=0.4mw ,L=180nm

Procedure

1.Design the CS amplifier as per the circuit diagram using LTSpice.

2.Set the components values as per the given parameters.

3.Create a new folder and save the LTSpice file in this folder along with the library file

4.Calculate the current value for the given power=0.4mw

a)Dc Analysis:

DC Analysis is done to check if the MOSFET is operating in the saturation region and to obtain the DC operation point.

procedure:

1.After setting up the circuit as per the circuit diagram, apply the dc voltage VDD=1.2V and Vgs=0.61

2.Go to Simulate option in the tab and the DC opt option and click 'ok'. 3.Click on run to run the circuit. 4.DC operating point is obtained.

Calculation(NMOS):

assume VRS=0.2V

Vov=VGS-Vth

Vgs=0.61

Vout=VDS+VRS

Vout=VDD/2+VRS

Vout=0.8v

P=V*I--(1)

I=0.2mA

Vg=Vgs+Vrs--(2)

Vg=0.61+0.2=0.81V

Vds>=Vgs-Vth--(3)

Vds>=0.25v--(it is in saturation state)

Vds=Id*Rs

Rs=1kohm

PMOS:

Vov=Vsg-|Vth|= VSGsg=0.64

Vs-Vg=0.64

Vs-0.64=Vg--(4) (Vs=VDD=1.2V)

Vg=0.56V

Id(PMOS)=Kn'w(Vsg-|Vth|)^2/2*L ----(5)

Tox=4.1*10^-9

Eox=3.98.85410^-12

U0=115.689*10^-4

solving 5th equation , we get W as,

W=11.823Um

ID(NMOS)=Kn'w(Vgs-Vth)^2/2*L ----(6)

Tox=4.1*10^-9

Eox=3.98.85410^-12

UO=273.809*10^-4

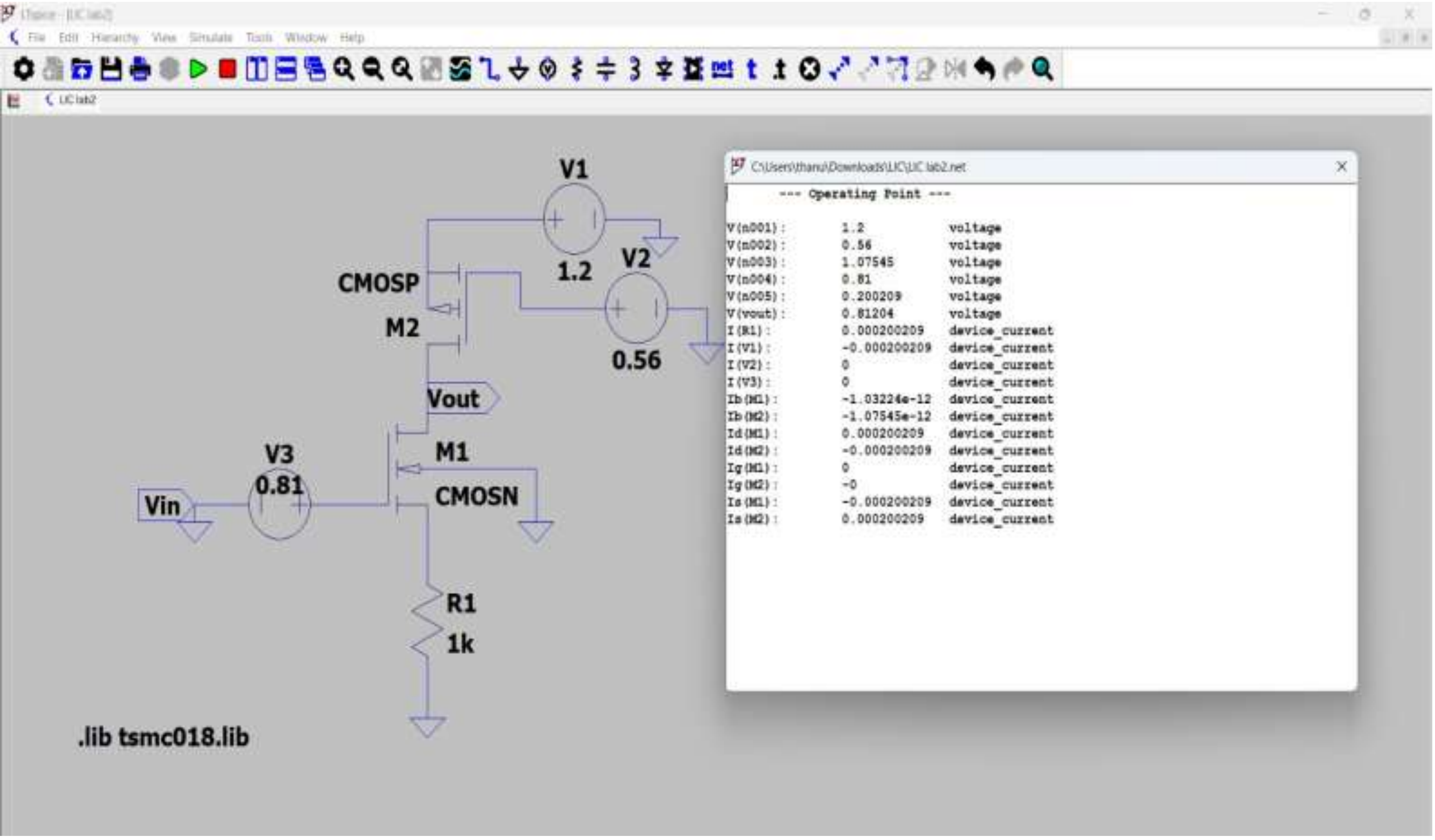
from 6th equation v will gwt W as

W=4.997*10^-6

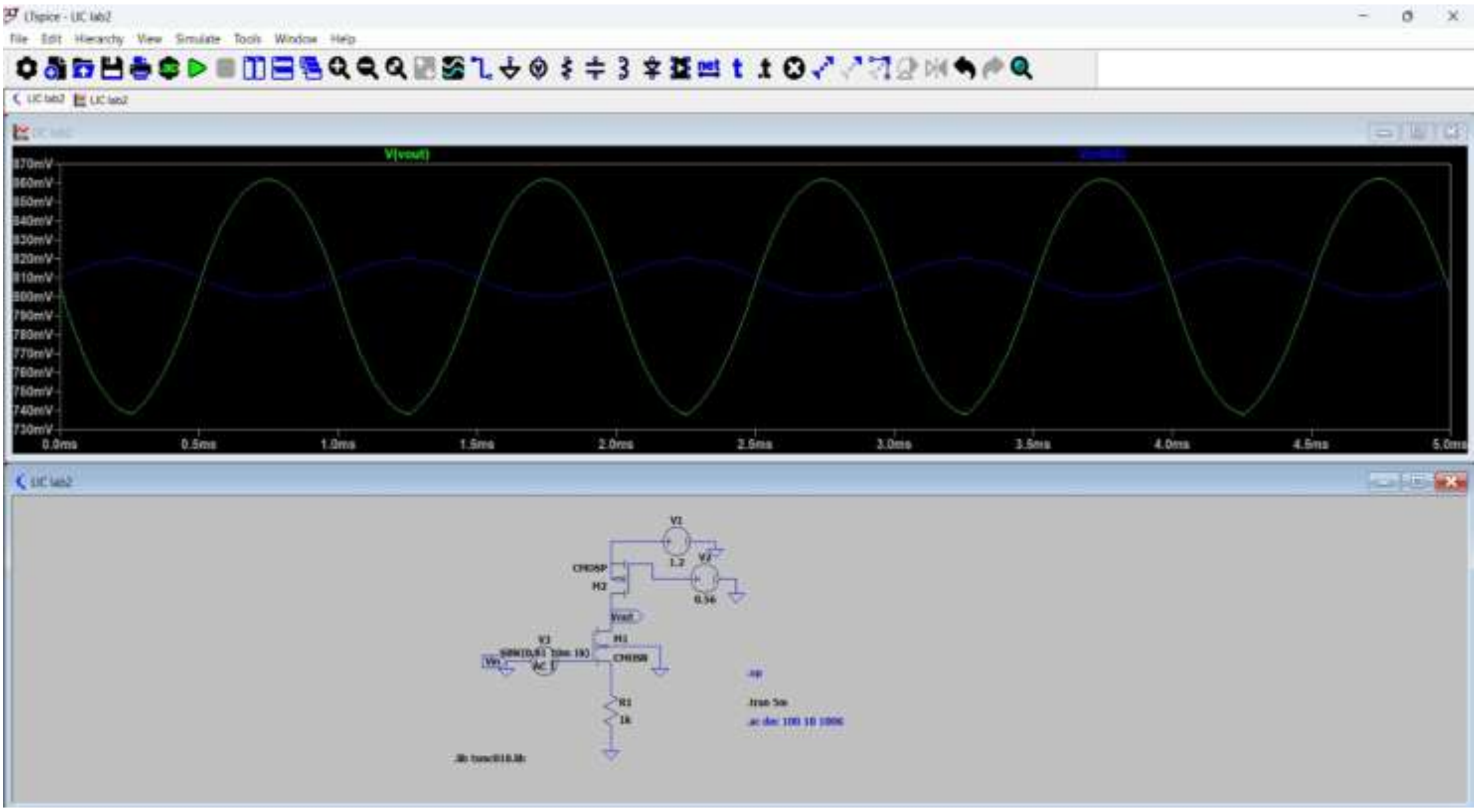
if we substitute these values in simulation we get,

Id has 100uA ,to get ID has 200uA, we have to increase the width of PMOS and NMOS

Wpmos=58.823u Wnmos=25.9954u for these values we will get Id has 200uA



b)Transient Analysis:



Input:

SINE(0.81 10m 1k)

10m = Peak amplitude

Vin(p-p) = 20 mV

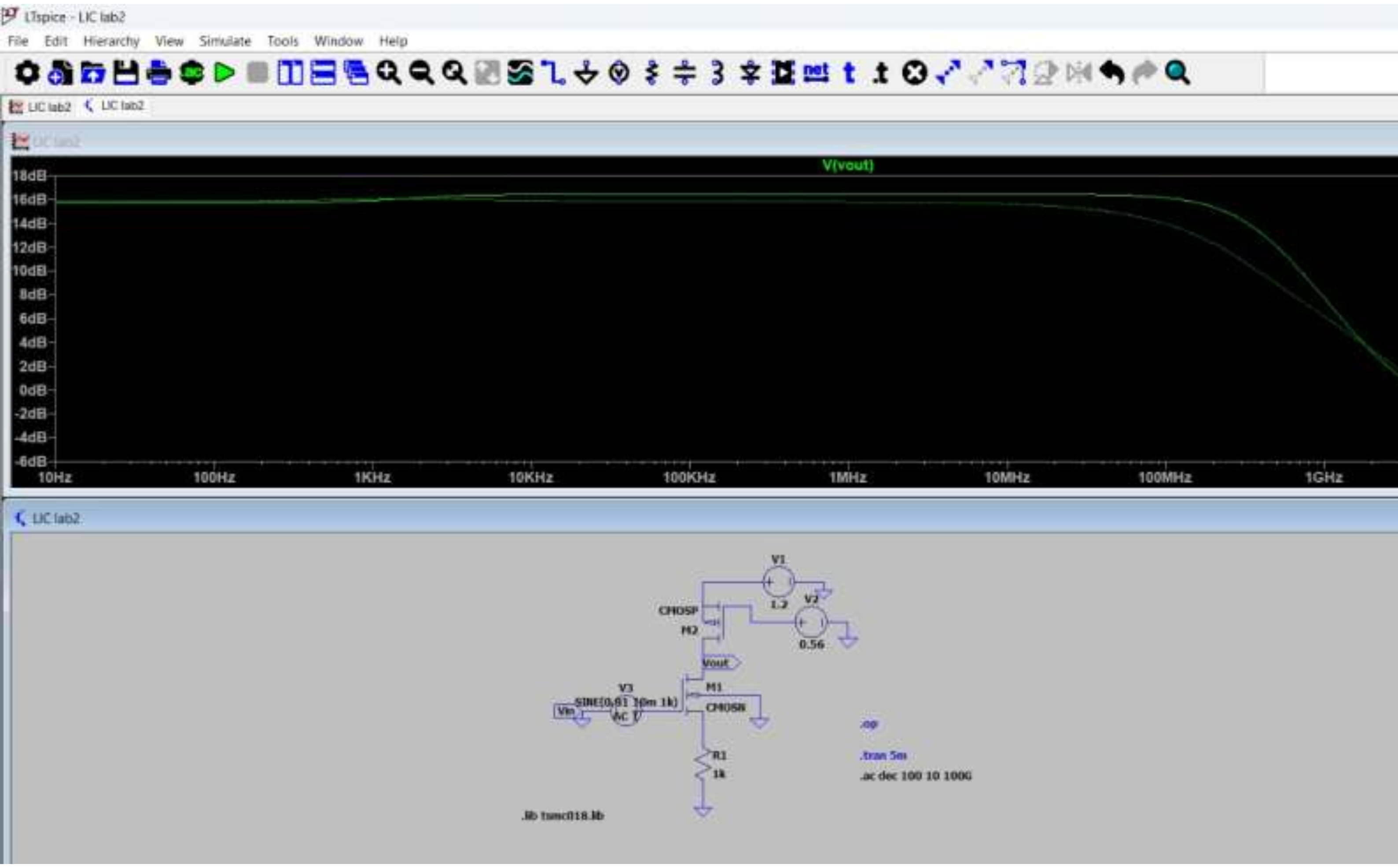
-Measured Vout ≈ 123 mV (approx)

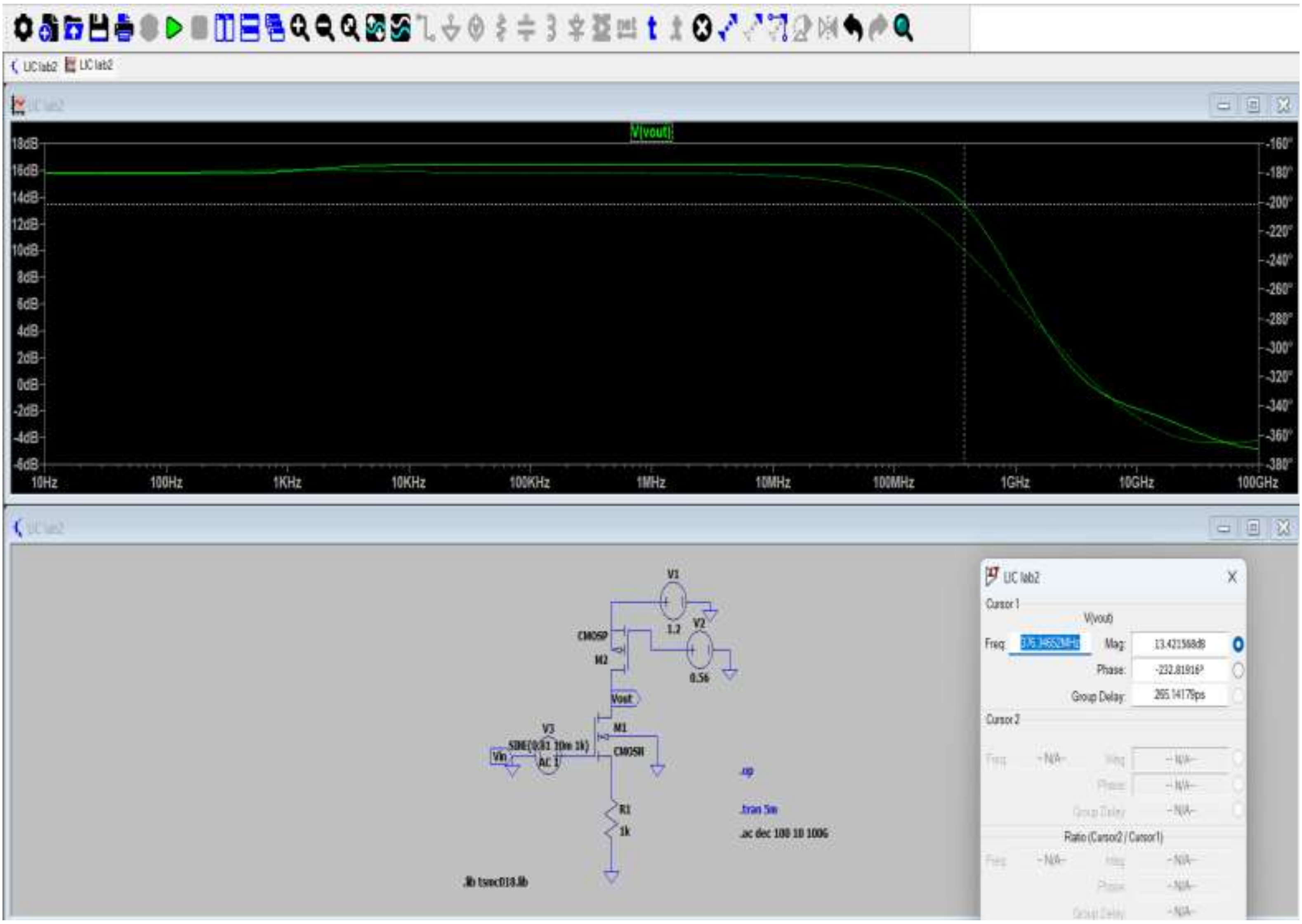
-Av = Vout(p-p) / Vin(p-p)

Av ≈ 6.30 V/V

-Av(dB) ≈ 16 dB

c)AC Analysis:





gm = 2Id/ Vov

gm = (2 × 200μA) / 0.25

gm = 1.6 mS

Assume λ = 0.1

ro = 1 / (λ Id)

ro = 1 / (0.1 × 200μA)

r_o = 50 kΩ

(r_{o_n} || r_{o_p}) ≈ 25 kΩ

$A_v = -g_m (r_{o\_n} \| r_{o\_p}) / (1 + g_m R_s)$

$A_v = (-1.6\text{mS} \times 25\text{k}\Omega) / (1 + (1.6\text{mS} \times 1\text{k}\Omega))$

$A_v \approx -15.38$ V/V

Av(dB) = 20 log(15.38)

Av(dB) = 23.74 dB

High cutoff frequency (fH) = 376.34 MHz

Low cutoff frequency (fL) ≈ 0 Hz

Bandwidth: BW = 376.34 MHz

Comparison:

Theoretical Gain = 23.74 dB

Simulated Gain = 16 dB

Theoretical Av = 15.38 V/V

Simulated Av = 6.30 V/V

Bandwidth = 376 MHz

Inference:

Simulated gain is lower due to:

Channel length modulation

Source degeneration resistance

Non-ideal MOS effects

AC gain differs from transient gain due to small signal assumption in AC analysis.

result:

amplifier theoretically achieves 23.74 dB gain with 376 MHz bandwidth, but in simulation, the gain drops to 16 dB (6.3 V/V) due to practical MOSFET non-idealities.